Week 2 Assignment Solutions

4. Library ISBN Normalizer & Validator

Normalize the publisher code and validate the 13-character format: 3 letters followed by 10 digits.

Java Solution

```
public class Main {
    static String normalizeCode(String raw) {
        String code = raw.trim();
        if (code.length() < 3) {
            return code.toUpperCase(); // lets validation report wrong length
        }
        return code.substring(0, 3).toUpperCase() + code.substring(3);
    }

    static String validateAndFormat(String code) {
        if (code.length() != 13) {
            return "Invalid: code must be exactly 13 characters";
        }
        for (int i = 0; i < 3; i++) {
            if (!Character.isLetter(code.charAt(i))) {
                return "Invalid: publisher code must be 3 letters";
            }
        }
        for (int i = 3; i < 13; i++) {
            if (!Character.isDigit(code.charAt(i))) {
                return "Invalid: body must contain only digits";
            }
        }

        StringBuilder result = new StringBuilder();
        result.append("[").append(code.substring(0, 3)).append("] YEAR: ")
              .append(code.substring(3, 7)).append(" | CATALOG: ")
              .append(code.substring(7));
        return result.toString();
    }

    public static void main(String[] args) {
        String code = normalizeCode(" pen2026004251 ");
        System.out.println(validateAndFormat(code));
        System.out.println(validateAndFormat(normalizeCode("12N2026004251")));
    }
}
```

Sample Output

```
[PEN] YEAR: 2026 | CATALOG: 004251
Invalid: publisher code must be 3 letters
```

Notes

Validation is staged: length first, publisher letters second, and the remaining ten digits last. No regular expressions are used.